REMEDIATION & GROWTH PLAN

Issues, Fixes & 30-Day Plan

taptoglass.com

Every material finding, numbered and prioritised, each paired with a ready-to-implement fix — exact configuration values, verbatim policy copy, and drop-in code. Followed by a day-by-day 30-day plan that builds topical authority without compromising the property's privacy-first, lightweight character.

31	6	14
ISSUES LOGGED	CRITICAL	QUICK WINS

Audit date: 29 August 2026

Companion to Document 1

Document 2 of 2

A How to Use This Document

Issues are numbered 1–31 and ordered by severity, then by implementation cost. Numbering is stable — cite it in tickets and commits.

LABEL	MEANING
CRITICAL	Blocks launch, revenue, or indexing; or creates legal exposure. Fix before promoting the site.
HIGH	Materially degrades trust, discoverability, or conversion. Fix within 30 days.
MEDIUM	Real defect with bounded impact. Schedule deliberately.
LOW	Polish. Batch opportunistically.
QUICK WIN	Under ~60 minutes, low risk, no dependencies.

Two issues must be verified against the live origin before fixing

Issues **1** and **13** concern values that this audit could measure only in a local build (no egress — see Document 1 §00). Confirm the live behaviour first; if already correct, close them and note the verification date.

Severity distribution

SEVERITY	COUNT	ISSUE NUMBERS
CRITICAL	6	1, 2, 3, 4, 5, 6
HIGH	9	7–15
MEDIUM	10	16–25
LOW	6	26–31

B Critical Issues (1–6)

01 Canonical origin defaults to the wrong domain

CRITICAL

File: web/lib/site.ts **Effort:** 5 min **Impact:** Total loss of indexing attribution QUICK WIN

`SITE_URL` falls back to `https://waterqualitylens.com`. No `.env`, CI workflow, or deploy config in the repository sets `NEXT_PUBLIC_SITE_URL`. If it is unset in production, every canonical tag, `og:url`, sitemap entry, JSON-LD `@id`, and the `robots.txt` `Host`/`Sitemap` directives will name a different domain — telling Google the real content lives elsewhere.

STEP 1 — VERIFY LIVE (DO THIS FIRST)

```
curl -s https://taptoglass.com/robots.txt
# Must print Host: https://taptoglass.com — NOT waterqualitylens.com

curl -s https://taptoglass.com/ | grep -o '<link rel="canonical"[^>]*>'
# Must print href="https://taptoglass.com"
```

STEP 2 — SET THE ENVIRONMENT VARIABLE IN THE HOST

```
# Vercel / Netlify / Cloudflare Pages → Environment Variables (all environments)
NEXT_PUBLIC_SITE_URL=https://taptoglass.com
```

STEP 3 — MAKE THE DEFAULT SAFE (VERBATIM REPLACEMENT)

```
// web/lib/site.ts — replace the whole file

/**
 * Single source of truth for the canonical origin. The production domain is the
 * default so a missing env var degrades to correct behaviour rather than
 * silently publishing another domain's canonicals.
 */
export const SITE_URL = (
  process.env.NEXT_PUBLIC_SITE_URL ?? "https://taptoglass.com"
).replace(/\/$/, "");

export const SITE_NAME = "Tap to Glass";
```

02 No privacy policy, terms, or contact method

CRITICAL

Files: new /privacy, /terms, /contact **Effort:** 1 day **Impact:** Blocks AdSense & affiliate programmes; YMYL E-E-A-T penalty

`/legal` is a self-described summary that "does not replace formal terms." No contact address exists anywhere in the repository. For health-adjacent YMYL content, identifiable ownership and contactability are core trust signals.

VERBATIM PRIVACY POLICY — READY TO PUBLISH AT /PRIVACY

Privacy Policy

Last updated: [DATE]

Tap to Glass ("we", "us") operates taptoglass.com. This policy explains what we collect, what we do not, and who else sees your data. It is written to be read.

The short version. We have no advertising, no analytics, no tracking pixels, and no third-party scripts. We set no cookies. You do not need an account to use the lookup.

Information you provide. When you use the address lookup, you enter a street address or ZIP code and optionally a dwelling type. We use these only to identify your public water system and generate your report. We do not require your name, email address, or phone number, and we do not sell, rent, or share your address with advertisers or data brokers.

Third parties that receive your address. To resolve an address to a water system, we transmit what you enter to two United States government services: the *U.S. Census Bureau Geocoding Service* (geocoding.geo.census.gov), which converts the address to coordinates, and the *U.S. EPA Envirofacts* service (data.epa.gov), which returns public water system records. Both are federal government systems operating under their own published privacy practices. We send no other data to any third party.

Information collected automatically. Our hosting provider records standard server logs, which may include IP address, user agent, and requested URL, for security and reliability. These logs are retained for no more than 30 days and are not combined with lookup addresses to build a profile.

Storage on your device. We store a single item in your browser's local storage, named `theme`, holding the value "light" or "dark" so the site remembers your display preference. It contains no identifier, is never transmitted to us, and you can clear it at any time through your browser's site-data settings. We use no cookies.

Affiliate links. Some hardware and laboratory-test links are affiliate placements, marked "Verified Partner." If you follow one, the destination retailer or laboratory operates under its own privacy practices and may set its own cookies. We do not control and are not responsible for those practices.

Your rights. Because the core tool is anonymous, we typically hold no personal information tied to you. If you believe we hold information about you, you may request access, correction, or deletion by emailing **[CONTACT EMAIL]**. Residents of California (CCPA/CPRA) and the European Economic Area and United Kingdom (UK GDPR/GDPR) may exercise the rights available under those laws through the same address, and we will respond within the period the applicable law requires. We do not sell or share personal information as those terms are defined under the CPRA.

Children. This service is not directed to children under 13 and we do not knowingly collect information from them.

Changes. Material changes will be reflected here with an updated date above.

Contact. [CONTACT EMAIL] · [LEGAL ENTITY NAME] · [MAILING ADDRESS]

VERBATIM TERMS OF SERVICE — READY TO PUBLISH AT /TERMS

Terms of Use

Last updated: [DATE]

What this service is. Tap to Glass is an informational tool that aggregates public municipal water data and independent product certifications. It is provided **as is** and **as available**, for general information only.

Not professional advice. Nothing on this site is medical, legal, engineering, or professional advice. We report engineering measurements and hardware certifications. We do not diagnose exposure, predict health outcomes, or recommend treatment. Consult a licensed professional for questions about your health.

Data accuracy. Contaminant and violation data originates from the U.S. EPA and state primacy agencies. Water system boundaries are derived from public and modeled datasets. We present this information faithfully but cannot guarantee it is current, complete, or free of errors in the source. Water quality changes between sampling events, and system-level data cannot describe contamination introduced by your own plumbing.

Verify before acting. Confirm the utility name on your water bill, particularly where we label a match "Modeled" or otherwise low-confidence, and obtain a certified laboratory test before making consequential decisions.

Affiliate relationships. We earn commissions on some hardware and laboratory-test links. Products qualify for display only by holding the relevant NSF/ANSI health certification for the contaminants detected; commercial arrangements never alter eligibility or ranking. See our Legal & Disclosures page.

Acceptable use. Use the service for personal, non-commercial research. Do not scrape, bulk-download, or resell the data, and do not use automated systems to submit requests at a rate that degrades the service.

Limitation of liability. To the maximum extent permitted by law, Tap to Glass and its operators are not liable for any indirect, incidental, or consequential damages arising from use of this service or reliance on its data.

Governing law. These terms are governed by the laws of [STATE/JURISDICTION].

Contact. [CONTACT EMAIL]

CONTACT PAGE — MINIMUM VIABLE CONTENT

Contact

General enquiries and corrections: [CONTACT EMAIL]

Data corrections. If a water system record or filter certification on this site is wrong, tell us. Include the PWSID or SKU and the source you are citing, and we will verify it against the primary record and correct it.

Privacy requests: [CONTACT EMAIL]

Operated by: [LEGAL ENTITY NAME], [MAILING ADDRESS]

We aim to respond within five business days.

WIRE INTO THE FOOTER

```
// web/components/site/SiteFooter.tsx — in the "Company" group
{ href: "/methodology", label: "Methodology" },
{ href: "/about",       label: "About" },
{ href: "/contact",     label: "Contact" },
{ href: "/legal",       label: "Legal & disclosures" },
{ href: "/privacy",     label: "Privacy policy" },
{ href: "/terms",       label: "Terms of use" },
```

Also add all three routes to `web/app/sitemap.ts` with `changeFrequency: "yearly", priority: 0.3`.

03 Data coverage limited to three water systems

CRITICAL

Files: web/lib/data.ts, src/ingest/sdwisIngest.ts, wrangler.toml **Effort:** Multi-week **Impact:** Blocks all revenue

The root commercial defect. `DEMO_SYSTEMS` contains three hand-written entries. Outside them the ZIP fallback returns `detections: []`, so `matchFilters()` returns an empty array and no hardware — therefore no affiliate revenue — is ever shown. See Document 1 §08.

STEP 1 — PROVISION THE CLOUDFLARE BINDINGS

```
wrangler d1 create waterlens
wrangler kv namespace create CACHE
wrangler queues create waterlens-alerts
wrangler queues create waterlens-alerts-dlq

# Paste the returned ids into wrangler.toml, replacing:
database_id = "REPLACE_WITH_D1_DATABASE_ID"
id          = "REPLACE_WITH_KV_NAMESPACE_ID"
```

STEP 2 — MIGRATE, SEED, DEPLOY, INGEST

```
npm run db:migrate:remote
wrangler secret put JWT_SECRET
npm run deploy
curl -X POST https://<worker>/api/admin/ingest -H "authorization: Bearer $JWT_SECRET"
```

STEP 3 — POINT THE WEB APP AT THE DEPLOYED API

```
// Replace the bundled-data path in web/lib/lookup.ts with a call to the
// deployed Worker, so one engine serves both surfaces and the duplicated
// logic in web/lib/{engine,geocode,lookup}.ts can be retired (Issue 16).
NEXT_PUBLIC_API_BASE=https://api.taptoglass.com
```

> **Interim mitigation — ship this week regardless**
>
> Until national ingestion runs, the site must not imply coverage it lacks. Implement Issue 4 (coverage honesty) as the stopgap. A tool that says "we don't have your system yet, here's what to do instead" retains trust; one that silently returns nothing destroys it.

04 Out-of-coverage results are indistinguishable from clean water

CRITICAL

Files: web/components/results/EmptyState.tsx, Hero.tsx **Effort:** 2–3 hrs **Impact:** Direct trust damage QUICK WIN

A user outside the three bundled systems receives a report with no contaminants and no filters. Nothing distinguishes "we have no data for your system" from "your water is clean." For a trust-first brand this is the most damaging possible failure mode, and it is fixable today without waiting on Issue 3.

VERBATIM COPY — THE "NO DATA" STATE (MUST NOT READ AS REASSURANCE)

> **We don't have sampling data for this water system yet**
>
> We identified your supplier as **{utilityName}**, but we do not yet hold contaminant sampling records for it. **This is a gap in our coverage — it is not a finding that your water is clean.** We cannot tell you either way, and we will not guess.
>
> **What you can do right now:**
>
> - **Read your utility's Consumer Confidence Report.** Every community water system must publish one annually by 1 July. Search "{utilityName} consumer confidence report" or request it directly — they are required to provide it.
> - **Check the EPA's records directly.** Your system's federal identifier is **{pwsid}**. Violation history is public via EPA Envirofacts under that ID.
> - **Test at the tap.** A certified laboratory test is the only method that measures the water you actually drink, including anything your own plumbing contributes. It is also the only way to detect a lead service line.
>
> We are expanding coverage system by system. [Tell us to prioritise this one →]

VERBATIM COPY — SET EXPECTATIONS BEFORE THE ASK (HERO SUB-LINE)

> Full contaminant reports are currently available for selected water systems, with national coverage rolling out. Every address returns its verified supplier and federal violation history.

Place directly beneath the search field. Honest, specific, and does not undersell what the tool does deliver.

05 Brand identity contradicts the domain on every page

CRITICAL

Files: site.ts, layout.tsx, manifest.ts, SiteFooter.tsx, data.ts, opengraph-image.tsx **Effort:** 3–4 hrs

Impact: Brand incoherence; SERP/social mismatch

Every title, the application name, the manifest, the JSON-LD Organization node, the copyright line, the OG share card and the affiliate redirect subdomain read "WaterQualityLens." The domain is taptoglass.com.

> **Decide first, then execute**
>
> Either (a) rename the product to **Tap to Glass** everywhere, or (b) move the site to waterqualitylens.com. Both are defensible; the present split is not. The fixes below assume (a) — the domain is the asset already purchased, and "Tap to Glass" is the more memorable consumer name.

GLOBAL RENAME — EXACT STRING REPLACEMENTS

```
# web/lib/site.ts
export const SITE_NAME = "Tap to Glass";

# web/app/layout.tsx — metadata
title: {
  default: "Tap to Glass — Address-level tap water quality intelligence",
  template: "%s · Tap to Glass",
},

# web/app/manifest.ts
name:       "Tap to Glass — address-level tap water quality intelligence",
short_name: "Tap to Glass",

# web/components/site/SiteFooter.tsx
© {new Date().getFullYear()} Tap to Glass. All rights reserved.

# web/app/page.tsx — FAQ entity names
"How does WaterQualityLens find my water system?"
"How does Tap to Glass find my water system?"
"Does WaterQualityLens make medical claims?"
"Does Tap to Glass make medical claims?"

# web/lib/data.ts — all 11 affiliate URLs
https://affil.waterqualitylens.com/...
https://go.taptoglass.com/...

# web/app/opengraph-image.tsx and apple-icon.tsx — wordmark text
```

Verify with `grep -ril "waterqualitylens" web/ src/` — must return nothing but this audit's own files.

`06` Undisclosed transmission of user addresses to third parties

CRITICAL

Files: web/lib/geocode.ts, new /privacy **Effort:** 30 min **Impact:** Compliance exposure QUICK WIN

Every address entered is transmitted to `geocoding.geo.census.gov`, and ZIP-fallback lookups additionally query `data.epa.gov`. The privacy summary states only that an address "is not sold" — users are not told it leaves the site at all. Both recipients are US government services, so the design is benign; the non-disclosure is the defect.

COVERED BY THE POLICY IN ISSUE 2 — PLUS IN-CONTEXT DISCLOSURE

> Your address is sent to the U.S. Census Bureau geocoder to locate it, and to the U.S. EPA to retrieve your water system's records. It is not stored, profiled, or sold. How we handle your data →

Render beneath the search field at `text-xs` in `ink-500`. Disclosure at the point of collection is materially stronger than disclosure in a linked policy, and it reinforces rather than undermines the privacy-first positioning.

C High-Priority Issues (7–15)

07 PWA manifest declares an icon path that does not exist

HIGH

File: web/app/manifest.ts **Effort:** 20 min **Impact:** 404 on every install attempt; not installable QUICK WIN

Verified against the build's own route manifest: the only generated icon route is `/apple-icon`. The manifest declares `/apple-icon.png`, which 404s. Separately, Chrome requires 192×192 and 512×512 PNG icons for install eligibility — neither exists, so the site is not installable despite declaring `display: standalone`.

VERBATIM REPLACEMENT — WEB/APP/MANIFEST.TS

```
import type { MetadataRoute } from "next";

export default function manifest(): MetadataRoute.Manifest {
  return {
    id: "/",
    name: "Tap to Glass — address-level tap water quality intelligence",
    short_name: "Tap to Glass",
    description:
      "Identify the public water system serving a US address, review the " +
      "contaminants it reports against federal limits with full provenance, " +
      "and match the retail filters independently NSF/ANSI-certified to remove them.",
    start_url: "/",
    scope: "/",
    display: "standalone",
    orientation: "portrait-primary",
    background_color: "#f6f7f9",
    theme_color: "#1b3e59",
    categories: ["utilities", "reference", "health"],
    icons: [
      { src: "/icon.svg",      sizes: "any",     type: "image/svg+xml", purpose: "any" },
      { src: "/icon-192.png",  sizes: "192x192", type: "image/png",     purpose: "any" },
      { src: "/icon-512.png",  sizes: "512x512", type: "image/png",     purpose: "any" },
      { src: "/icon-maskable-512.png", sizes: "512x512", type: "image/png", purpose: "maskable" },
      { src: "/apple-icon.png", sizes: "180x180", type: "image/png" },  ← DELETE: 404s
    ],
  };
}
```

Place the three PNGs in `web/public/`. The maskable variant needs ~20% safe-zone padding so Android's circular mask does not clip the mark. The HTML `<link rel="apple-touch-icon">` is already correct and needs no change.

08 No rate limiting on any API route

HIGH

Files: web/app/api/lookup/route.ts, src/router.ts **Effort:** 2 hrs **Impact:** Abuse vector; risk of Census IP block

`/api/lookup` accepts unbounded requests, each able to trigger an outbound call to a no-SLA government geocoder. Sustained abuse risks having the origin blocked by Census, which would take down lookups entirely.

DROP-IN EDGE RATE LIMITER — WEB/MIDDLEWARE.TS (NEW FILE)

```
import { NextResponse, type NextRequest } from "next/server";

// Fixed-window limiter. In-memory per isolate — adequate as a first line of
// defence; move to KV/Upstash when traffic justifies shared state.
const WINDOW_MS = 60_000;
const MAX_REQUESTS = 20;
const hits = new Map<string, { count: number; resetAt: number }>();

export function middleware(req: NextRequest) {
  const ip =
    req.headers.get("cf-connecting-ip") ??
    req.headers.get("x-forwarded-for")?.split(",")[0]?.trim() ??
    "unknown";

  const now = Date.now();
  const entry = hits.get(ip);

  if (!entry || now > entry.resetAt) {
    hits.set(ip, { count: 1, resetAt: now + WINDOW_MS });
  } else if (++entry.count > MAX_REQUESTS) {
    return new NextResponse(
      JSON.stringify({ error: "Too many requests. Please wait a minute." }),
      {
        status: 429,
        headers: {
          "content-type": "application/json",
          "retry-after": String(Math.ceil((entry.resetAt - now) / 1000)),
        },
      }
    );
  }

  // Opportunistic sweep so the map cannot grow without bound.
  if (hits.size > 10_000) {
    for (const [k, v] of hits) if (now > v.resetAt) hits.delete(k);
  }

  return NextResponse.next();
}

export const config = { matcher: ["/api/:path*", "/results"] };
```

09 Entity pages carry no structured data beyond breadcrumbs

HIGH

Files: app/pws/[pwsid]/page.tsx, app/contaminants/[code]/page.tsx, app/registry/page.tsx **Effort:** 3 hrs

Impact: Lost rich results on the pages built to rank

The programmatic entity pages — the site's entire organic growth thesis — emit only `BreadcrumbList`. No `Dataset`, `DefinedTerm`, or `ItemList` markup exists.

PWS PAGES — ADD TO APP/PWS/[PWSID]/PAGE.TSX

```
const pwsJsonLd = {
  "@context": "https://schema.org",
  "@type": "Dataset",
  "@id": `${SITE_URL}/pws/${u.pwsid}#dataset`,
  name: `${u.name} — drinking water quality data (${u.pwsid})`,
  description:
    `Contaminant sampling results, federal violation history and NSF/ANSI-certified ` +
    `filtration matches for ${u.name}, serving ${u.population_served?.toLocaleString()} ` +
    `people in ${u.county} County, ${u.state}.`,
  url: `${SITE_URL}/pws/${u.pwsid}`,
  license: "https://www.epa.gov/web-policies-and-procedures/epa-web-site-content-policies",
  isAccessibleForFree: true,
  creator: { "@type": "Organization", name: "U.S. Environmental Protection Agency" },
  publisher: { "@id": `${SITE_URL}/#organization` },
  temporalCoverage: profile.history.at(0)?.sample_date + "/" + profile.history.at(-1)?.sample_date,
  spatialCoverage: {
    "@type": "Place",
    name: `${u.county} County, ${u.state}`,
    address: { "@type": "PostalAddress", addressRegion: u.state, addressCountry: "US" },
  },
  variableMeasured: profile.detections.map((d) => ({
    "@type": "PropertyValue",
    name: d.name,
    value: d.value,
    unitText: d.unit,
    maxValue: d.mcl ?? undefined,
    measurementTechnique: "EPA SDWIS compliance sampling",
  })),
};
```

CONTAMINANT PAGES — ADD TO APP/CONTAMINANTS/[CODE]/PAGE.TSX

```
const contaminantJsonLd = {
  "@context": "https://schema.org",
  "@type": ["DefinedTerm", "MedicalWebPage"],
  "@id": `${SITE_URL}/contaminants/${c.code}#term`,
  name: c.name,
  termCode: c.code,
  description: c.health_effects,
  url: `${SITE_URL}/contaminants/${c.code}`,
  inDefinedTermSet: {
    "@type": "DefinedTermSet",
    name: "EPA regulated drinking water contaminants",
    url: `${SITE_URL}/contaminants`,
  },
  // Keeps the page inside the medical-information vertical without
```

```
  // making a treatment claim — consistent with the site's medical disclaimer.
  medicalAudience: "Patient",
  lastReviewed: new Date().toISOString().slice(0, 10),
};
```

REGISTRY — ADD TO APP/REGISTRY/PAGE.TSX

```
const registryJsonLd = {
  "@context": "https://schema.org",
  "@type": "ItemList",
  name: "Filter Certification Registry",
  description:
    "Retail water filters audited against their NSF/ANSI certifications, " +
    "including SKUs marketed on aesthetic-only standards with no health certification.",
  numberOfItems: FILTERS.length,
  itemListElement: FILTERS.map((f, i) => ({
    "@type": "ListItem",
    position: i + 1,
    item: {
      "@type": "Product",
      name: `${f.brand} ${f.model}`,
      sku: f.sku,
      brand: { "@type": "Brand", name: f.brand },
      offers: { "@type": "Offer", price: f.price_usd, priceCurrency: "USD" },
    },
  })),
};
```

10 No author, credentials, or named publisher on YMYL content

HIGH

Files: app/about/page.tsx, app/methodology/page.tsx **Effort:** 2 hrs **Impact:** Largest single E-E-A-T deficit

Water quality is squarely "Your Money or Your Life." Google's quality guidance weights identifiable expertise and accountability heavily for such topics. The site currently publishes health-adjacent guidance with no named human or entity behind it.

VERBATIM ABOUT-PAGE SECTION — ADAPT BRACKETED FIELDS HONESTLY

> **Who is behind Tap to Glass**
>
> Tap to Glass is built and maintained by **[NAME / ENTITY]**, [one sentence of relevant background — data engineering, environmental science, public-health informatics, or similar]. We are not affiliated with any water utility, filter manufacturer, or testing laboratory.
>
> **How we make money.** We earn affiliate commissions when readers purchase filtration hardware or laboratory tests through our links. We do not accept payment for placement, and no manufacturer can buy eligibility: a product appears only if it holds an independent NSF/ANSI health certification for the specific contaminants detected.
>
> **Our editorial standard.** Every figure we publish carries its source, sample date, and testing agency. Where our confidence in a result is low, we say so on the result itself rather than in a footnote. Where a filter is not certified for a contaminant, we exclude it rather than qualify it.
>
> **Corrections.** If something here is wrong, write to **[CONTACT EMAIL]** with the identifier and your source. We verify against the primary record and publish corrections.

ADD ORGANIZATION DETAIL TO THE SITE JSON-LD (LAYOUT.TSX)

```
{
  "@type": "Organization",
  "@id": `${SITE_URL}/#organization`,
  name: SITE_NAME,
  url: SITE_URL,
  logo: `${SITE_URL}/icon.svg`,
  founder: { "@type": "Person", name: "[NAME]" },
  email: "[CONTACT EMAIL]",
  knowsAbout: [
    "drinking water quality", "NSF/ANSI water filtration standards",
    "EPA Safe Drinking Water Act compliance", "PFAS remediation",
  ],
}
```

11 Lab-test affiliate opportunity is referenced but never actionable

HIGH

Files: components/results/Disclaimers.tsx, EmptyState.tsx **Effort:** 2 hrs **Impact:** Highest-yield unexploited revenue path

QUICK WIN

Two disclaimers correctly tell the user that only a point-of-use lab test can detect plumbing-borne contamination — and neither offers a way to get one. This is the single most valuable gap on the site: lab referrals pay materially better than filter hardware, and unlike hardware they are relevant for **every** address, including those with no data. This revenue path works despite the coverage gap rather than being blocked by it.

VERBATIM CTA — PLACE AFTER THE PLUMBING DISCLAIMER AND IN EVERY EMPTY STATE

> **Only a tap test can see your own plumbing**
>
> The data above describes water as it leaves your utility. It cannot see what your home's pipes add to it — and a legacy lead service line is invisible to every system-level record, including ours.
>
> A certified laboratory test on a sample from your own tap is the only way to close that gap. We partner with an accredited testing laboratory and earn a commission if you order through this link; we recommend testing regardless of where you buy it.
>
> [Order a certified tap test →] *Verified Partner*

Apply `rel="sponsored noopener noreferrer"`, as the existing `Button external` component already does.

12 CSP permits 'unsafe-inline' for scripts

HIGH

File: web/next.config.mjs **Effort:** 45 min **Impact:** Removes CSP's main XSS defence QUICK WIN

There is exactly one inline script on the site — the theme bootstrap — and its content is a fixed constant. That makes a SHA-256 hash the correct fix; no nonce plumbing is required.

STEP 1 — COMPUTE THE HASH OF THE EXACT SCRIPT BODY

```
# The digest must cover the script's text content byte-for-byte,
# with no surrounding whitespace. Recompute whenever THEME_BOOTSTRAP changes.
node -e 'const s=require("crypto").createHash("sha256");
s.update(process.argv[1]);
console.log("sha256-"+s.digest("base64"))' "$(node -p "
  require(\"fs\").readFileSync(\"app/layout.tsx\",\"utf8\")
    .match(/const THEME_BOOTSTRAP = \`([\s\S]*?)\`;/)[1]
")"
```

STEP 2 — REPLACE THE SCRIPT-SRC DIRECTIVE

```
// web/next.config.mjs
const THEME_SCRIPT_HASH = "'sha256-REPLACE_WITH_COMPUTED_DIGEST'";

const csp = [
  "default-src 'self'",
  "script-src 'self' 'unsafe-inline'",
  `script-src 'self' ${THEME_SCRIPT_HASH}`,
  "style-src 'self' 'unsafe-inline'",   // retained: Next/Tailwind inject style tags
  "img-src 'self' data: blob:",
  "font-src 'self'",
  "connect-src 'self'",
  "form-action 'self'",
  "frame-ancestors 'self'",
  "base-uri 'self'",
  "object-src 'none'",
  "report-to csp-endpoint",
  "upgrade-insecure-requests",
].join("; ");
```

Verify in DevTools that no CSP violation is logged and that dark mode still resolves before first paint. If Next emits additional inline bootstrap scripts in a future version, switch to a nonce via middleware.

13 Security headers unverified at the edge

HIGH

Scope: Production origin **Effort:** 15 min **Impact:** Header set may not be delivered QUICK WIN

Headers declared in `next.config.mjs` are emitted only when the app is served by the Next.js runtime. A static export, or a CDN in front of the origin, can drop or override them. This audit could not verify delivery.

VERIFICATION COMMAND

```
curl -sSI https://taptoglass.com/ | grep -iE \
  'strict-transport|content-security|x-content-type|referrer-policy|permissions-policy|x-frame|cross-
origin'

# Expected, in full:
strict-transport-security: max-age=63072000; includeSubDomains; preload
content-security-policy: default-src 'self'; script-src 'self' 'sha256-...'; ...
x-content-type-options: nosniff
referrer-policy: strict-origin-when-cross-origin
permissions-policy: geolocation=(), camera=(), microphone=(), payment=(), usb=()
x-frame-options: SAMEORIGIN
cross-origin-opener-policy: same-origin

# And confirm /results is excluded from indexing:
curl -sSI 'https://taptoglass.com/results?address=test+address' | grep -i x-robots-tag
# Expected: x-robots-tag: noindex, nofollow
```

If headers are missing, replicate them at the CDN layer (`_headers` for Cloudflare Pages / Netlify, or `vercel.json`). Once confirmed, submit the domain to hstspreload.org — the HSTS directive is already preload-eligible.

14 Worker API returns Access-Control-Allow-Origin: * on authenticated routes

HIGH

File: src/lib/http.ts **Effort:** 30 min **Impact:** Moat dataset readable by any origin QUICK WIN

`JSON_HEADERS` applies `access-control-allow-origin: *` to every response, including `/api/vault/*` and `/api/addresses`, with `authorization` in the allowed headers. Bearer-token auth limits classic CSRF, but the curated certification dataset the README calls the platform's defensive moat is served to any origin that asks.

VERBATIM REPLACEMENT — SRC/LIB/HTTP.TS HEADER BLOCK

```
// Public read endpoints may be embedded anywhere; authenticated and
// proprietary-dataset routes are restricted to first-party origins.
const ALLOWED_ORIGINS = new Set([
  "https://taptoglass.com",
  "https://www.taptoglass.com",
]);

const BASE_HEADERS = {
  "content-type": "application/json; charset=utf-8",
  "cache-control": "no-store",
  "vary": "Origin",
};

export function corsHeaders(req: Request, scope: "public" | "private" = "private") {
  const origin = req.headers.get("Origin");
  if (scope === "public" && !origin) return {};
  if (origin && ALLOWED_ORIGINS.has(origin)) {
    return {
      "access-control-allow-origin": origin,
      "access-control-allow-methods": "GET,POST,PUT,DELETE,OPTIONS",
      "access-control-allow-headers": "content-type,authorization",
      "access-control-allow-credentials": "true",
    };
  }
  return {};
}
```

Thread `req` through `json()` and `error()`, and mark only genuinely public endpoints (`/api/contaminants`, `/api/pws/:id`) as `"public"`.

15 Results page blocks LCP on a live third-party geocode

HIGH

File: web/app/results/page.tsx **Effort:** 2 hrs **Impact:** LCP directly coupled to upstream latency

`/results` is `force-dynamic` and awaits a Census geocode with an 8-second timeout inside the render path. A slow upstream becomes a slow LCP. A `loading.tsx` exists, which helps perceived latency but does not improve the metric.

STREAM THE SHELL, SUSPEND THE DATA

```
// web/app/results/page.tsx
import { Suspense } from "react";

export default function ResultsPage({ searchParams }: { searchParams?: SearchParams }) {
  const address  = firstParam(searchParams?.address).trim();
  const dwelling = parseDwelling(searchParams?.dwelling_type);

  if (address.length < MIN_ADDRESS_LENGTH) return <PromptState />;

  return (
    <div className="bg-surface-sunken">
      {/* Static shell paints immediately — this becomes the LCP element. */}
      <ReportHeader address={address} />
      <Suspense fallback={<ReportSkeleton />}>
        {/* @ts-expect-error async server component */}
        <ReportBody address={address} dwelling={dwelling} />
      </Suspense>
    </div>
  );
}

// Move the awaited runLookup() into ReportBody so only that subtree suspends.
```

Also reduce the geocode timeout from 8s to ~4s: beyond that the gazetteer fallback is a better user experience than continuing to wait.

D Medium-Priority Issues (16–25)

16 Domain logic duplicated across two backends

MEDIUM

Files: src/services/* vs web/lib/* **Effort:** 1–2 days

Geocoding, spatial resolution, exceedance math and set-cover matching are implemented twice. They have already drifted: `web/lib/engine.ts`'s `resolveSpatial` has no Tier-2 path, so the "Matched Proxy" tier the UI documents is unreachable on the web surface except via the ZIP fallback's hardcoded label.

FIX

```
# Once Issue 3 lands, retire the duplicates and call the Worker:
git rm web/lib/engine.ts web/lib/geocode.ts web/lib/gazetteer.ts web/lib/envirofacts.ts
# Keep web/lib/lookup.ts as a thin typed fetch client against NEXT_PUBLIC_API_BASE.
# Until then, port the Tier-2 branch into web/lib/engine.ts so the tiers match.
```

17 Zero frontend tests

MEDIUM

Scope: web/ **Effort:** 4 hrs

The backend has 45 passing tests. `web/` has none — including the re-implemented set-cover engine, whose failure mode is recommending a filter that does not remove a detected hazard. That is the highest-consequence bug the codebase can produce.

MINIMUM VIABLE SUITE

```
npm i -D vitest @vitejs/plugin-react

// web/lib/__tests__/engine.test.ts — the safety-critical assertions
describe("matchFilters", () => {
  it("never recommends an NSF 42-only filter for a health contaminant", () => {
    const dets = [detection("PB90", 27)];
    const out  = matchFilters(dets, "UNKNOWN");
    expect(out.every(f => f.certified_standards.some(s => ["53","58","401"].includes(s)))).toBe(true);
    expect(out.find(f => f.sku === "BR-OB03")).toBeUndefined();  // 42 only
    expect(out.find(f => f.sku === "PL-UNIV01")).toBeUndefined(); // 42 + 372 only
  });

  it("requires full coverage — never a partial match", () => {
    const dets = [detection("PB90", 27), detection("1040", 12)]; // lead + nitrate
    // Only RO (58) covers nitrate; carbon-only SKUs must be excluded entirely.
    for (const f of matchFilters(dets, "UNKNOWN")) {
      expect(f.neutralizes).toEqual(expect.arrayContaining(["PB90","1040"]));
    }
  });

  it("returns nothing when no contaminant exceeds its health goal", () => {
    expect(matchFilters([detection("CU90", 0.4)], "UNKNOWN")).toEqual([]);
  });
});
```

18 Mobile navigation panel has no focus trap

MEDIUM

File: components/site/MobileNav.tsx **Effort:** 45 min QUICK WIN

Escape-to-close and focus-return are correctly implemented, but focus is never moved into the open panel and can tab out of it into the page behind. This is the one clear WCAG gap in an otherwise strong accessibility implementation.

ADD TO MOBILENAV

```
const panelRef = useRef<HTMLDivElement>(null);

useEffect(() => {
  if (!open) return;
  const panel = panelRef.current;
  if (!panel) return;

  const focusables = panel.querySelectorAll<HTMLElement>(
    'a[href], button:not([disabled]), [tabindex]:not([tabindex="-1"])'
  );
  focusables[0]?.focus();

  function onKey(e: KeyboardEvent) {
    if (e.key === "Escape") { setOpen(false); buttonRef.current?.focus(); return; }
    if (e.key !== "Tab" || focusables.length === 0) return;
    const first = focusables[0];
    const last  = focusables[focusables.length - 1];
    if (e.shiftKey && document.activeElement === first) { e.preventDefault(); last.focus(); }
    else if (!e.shiftKey && document.activeElement === last) { e.preventDefault(); first.focus(); }
  }
  document.addEventListener("keydown", onKey);
  return () => document.removeEventListener("keydown", onKey);
}, [open]);

// then: <div id={panelId} ref={panelRef} ...>
```

19 No live-region announcement during lookup

MEDIUM

File: components/AddressSearch.tsx **Effort:** 20 min QUICK WIN

The submit button label changes to "Checking…" but screen-reader users receive no announcement that a report is loading or has arrived.

FIX

```
<p role="status" aria-live="polite" className="sr-only">
  {pending ? "Looking up your water system, please wait." : ""}
</p>
```

On `/results`, give the report container `tabIndex={-1}` and focus it on mount so keyboard users land on the answer rather than at the top of the document.

20 Dwelling selector uses toggle buttons for single-select

MEDIUM

File: components/AddressSearch.tsx **Effort:** 45 min

Three `aria-pressed` buttons in a `fieldset` convey "three independent toggles," not "choose one of three." Assistive technology cannot infer the mutual exclusivity.

FIX — NATIVE RADIOS, VISUALLY STYLED AS CHIPS

```
<fieldset>
  <legend className="...">Dwelling type — tailors the hardware guidance</legend>
  {DWELLINGS.map((d) => (
    <label key={d.value} className="... has-[:checked]:border-brand-300 has-[:checked]:bg-brand-50">
      <input
        type="radio"
        name="dwelling"
        value={d.value}
        checked={dwelling === d.value}
        onChange={() => setDwelling(d.value)}
        className="sr-only"
      />
      {d.label}
    </label>
  ))}
</fieldset>
```

Native radios restore arrow-key navigation and correct group semantics at no visual cost.

21 PBKDF2 iteration count below current guidance

MEDIUM

File: src/lib/auth.ts **Effort:** 10 min QUICK WIN

100,000 iterations against an OWASP recommendation of 600,000 for PBKDF2-HMAC-SHA256. The construction is otherwise correct — per-user random salt, constant-time comparison, parameters stored alongside the hash, which means existing hashes keep verifying after the change.

FIX

```
const PBKDF2_ITERATIONS = 100_000;
const PBKDF2_ITERATIONS = 600_000;

// verifyPassword() already reads the iteration count from the stored hash,
// so old credentials continue to verify. Optionally re-hash on next successful
// login to migrate users forward.
```

22 og:site_name lost and twitter:title generic on the homepage

MEDIUM

File: web/app/page.tsx **Effort:** 10 min QUICK WIN

Verified in the built HTML: the page-level `openGraph` object overrides the root without re-declaring `siteName`, so no `og:site_name` is emitted. `twitter:title` falls back to the bare brand string rather than the page title.

FIX

```
openGraph: {
  type: "website",
  siteName: SITE_NAME,
  locale: "en_US",
  title: "Tap to Glass — Address-level tap water quality intelligence",
  description: "...",
  url: "/",
},
twitter: {
  card: "summary_large_image",
  title: "Tap to Glass — What's in your tap water, and what removes it",
  description: "...",
},
```

23 No outbound citations to primary sources

MEDIUM

Files: methodology, contaminant, PWS pages **Effort:** 2 hrs

EPA, NSF, WQA and TEMM are named throughout but never linked. On a site whose entire positioning is data provenance, unlinked sourcing is conspicuous — and authoritative outbound links are a positive quality signal.

ADD A SOURCES BLOCK TO EACH CONTAMINANT PAGE

```
<section aria-labelledby="sources-h">
  <h2 id="sources-h">Primary sources</h2>
  <ul>
    <li><a href="https://www.epa.gov/ground-water-and-drinking-water/national-primary-drinking-water-
regulations"
           rel="noopener">EPA National Primary Drinking Water Regulations</a> — MCL and MCLG values</li>
    <li><a href="https://www.epa.gov/enviro/sdwis-search" rel="noopener">EPA SDWIS</a>
        — sampling results and violation history</li>
    <li><a href="https://www.nsf.org/testing/water" rel="noopener">NSF International</a>
        — certification listings verified {LAST_AUDIT_DATE}</li>
  </ul>
</section>
```

24 No freshness signals on entity pages

MEDIUM

Files: contaminant and PWS pages **Effort:** 1 hr QUICK WIN

Sitemap `lastModified` is data-driven and correct, but no user-visible or machine-readable freshness marker appears on the pages themselves. For regulated data this is both a trust and a ranking signal.

FIX

```
<p className="text-xs text-ink-500">
  Certification data last audited <time dateTime={LAST_AUDIT}>{formatDate(LAST_AUDIT)}</time>
  {" · "}Sampling data current to <time dateTime={newest}>{formatDate(newest)}</time>
</p>

// and in the page metadata:
other: { "article:modified_time": newestSampleDate }
```

25 CSS bundle is large relative to the surface area

MEDIUM

Measured: 58.4 kB uncompressed **Effort:** 1 hr

Large for a site of this size. Likely driven by the extensive custom shadow, radius and fluid-type scales plus the dark-variant expansion of every utility.

INVESTIGATE

```
npx @next/bundle-analyzer
# or inspect the emitted stylesheet directly:
grep -o '\.[a-z-]*{' .next/static/css/*.css | sort | uniq -c | sort -rn | head -40

# Likely wins:
#  - prune unused steps from the boxShadow / borderRadius / fontSize scales
#  - confirm content globs are not over-matching
#  - ensure the CDN serves brotli (58.4 kB → ~9 kB)
curl -sI -H 'Accept-Encoding: br' https://taptoglass.com/_next/static/css/*.css | grep -i content-encoding
```

E Low-Priority Issues (26–31)

26 Obsolete keywords meta tag emitted

LOW

File: app/layout.tsx QUICK WIN

Ignored by every major engine since 2009. Harmless but signals dated practice to anyone inspecting the source.

```
keywords: ["tap water quality", "NSF 53 filter", ...],   ← delete
```

27 robots.txt Host directive is non-standard

LOW

File: app/robots.ts QUICK WIN

`Host:` is a Yandex extension; Google and Bing ignore it. Harmless, but canonical tags already carry the signal.

```
host: SITE_URL,   ← optional removal
```

28 /pws index page is thin

LOW

Measured: 394 rendered words

Three entries and little explanatory content. Resolves naturally with Issue 3; until then add an explanation of what a PWSID is, how systems are classified (CWS / NTNCWS / TNCWS), and what source water types mean — content that will remain useful once the index grows.

29 No search or filtering on entity indexes

LOW

Files: /contaminants, /registry

27 contaminants and 11 SKUs are browsable today. Add client-side filtering before either list exceeds ~50 rows. Filter by contaminant class and by health-vs-aesthetic certification respectively.

30 Copyright year fixed at build time

LOW

File: SiteFooter.tsx QUICK WIN

`new Date().getFullYear()` in a server component is evaluated when the page is prerendered, so a statically built site will display a stale year until redeployed. Either accept it with a scheduled rebuild, or hardcode a founding year: `© 2026 Tap to Glass`.

31 No CSP violation reporting

LOW

File: next.config.mjs QUICK WIN

CSP failures are currently silent, so a policy regression would go unnoticed.

```
{ key: "Reporting-Endpoints", value: 'csp-endpoint="https://taptoglass.com/api/csp-report"' }
// plus "report-to csp-endpoint" in the policy (see Issue 12), and a
// route handler that logs and returns 204.
```

F Monetization & Affiliate Opportunity Map

Mapped to the portfolio's established style: disclosed affiliate placement, no display advertising on the utility surface, and no compromise of the certification-first ranking that constitutes the moat.

OPPORTUNITY	YIELD	EFFORT	NOTES & SEQUENCING
Certified lab tests Tap Score / SimpleLab	HIGHEST	Low	**Do this first.** Higher commission than hardware, and relevant for *every* address — including the ~99.9% with no coverage. The plumbing disclaimer already argues the case honestly; it simply has no CTA. See Issue 11.
Filter hardware 11 SKUs live	High	Blocked	Infrastructure complete and correctly disclosed. Inert until coverage lands (Issue 3). Migrate the redirect domain to `go.taptoglass.com` (Issue 5).
Replacement cartridges	Recurring	Medium	The genuinely underrated play. Cartridges are consumable on a 3–6 month cycle against a one-time appliance purchase. Capture install date, email a replacement reminder, attach the affiliate link. Highest lifetime value per acquired user on the site.
Email list Water-quality alerts	Compounding	Medium	The Worker already implements saved addresses and alert delivery; none of it is exposed in the web UI. "Email me if my utility reports a violation" is a genuinely useful, privacy-defensible opt-in — and the only owned audience channel available.
Whole-home / softeners	High ticket	Medium	Only one whole-house SKU is stocked. Hard-water and sediment queries convert at high order values and are adjacent to existing content.
Display advertising	Low	Blocked	**Recommend against on the tool surface.** Blocked until Issue 2 anyway. Ad tech would forfeit the 87.1 kB bundle, the enforceable CSP, and the zero-tracking claim — the three assets that differentiate the property.
Utility/municipal licensing	Speculative	High	Longer-term. Small utilities must publish annual CCRs and many do it badly. The report template already produces a superior artifact.

> **Sequencing principle**
>
> Lab tests before hardware. Hardware requires coverage; lab tests do not, they pay better, and they are recommended at precisely the moment the site's own disclaimer establishes the need. It is the one revenue path that works *because* of the coverage gap rather than in spite of it — and it is honest, which is why it fits the brand.

Content-to-revenue mapping

CONTENT ASSET	INTENT	MONETIZATION PATH
/contaminants/PB90 (Lead)	Informational, high volume	Lab test → NSF 53 pitcher → under-sink RO
/contaminants/PFOA, PFOS	Informational, rising	NSF 53/58 only — highest-margin category
/contaminants/1040 (Nitrate)	Rural, high intent	RO exclusively — the nitrate warning already makes the technical case
/registry (deceptive SKUs)	Commercial investigation	Highest trust-to-conversion ratio on the site
/pws/[pwsid] entity pages	Local, transactional	The volume engine once Issue 3 lands
"How to read your CCR" (new)	Informational	Lab test; strong internal-link hub

G 30-Day Improvement & Content Plan

Sequenced so that trust and compliance land before traffic acquisition — there is no value in ranking a site that cannot be contacted, and active harm in ranking one whose canonicals name another domain. Content targets are additive to the existing 27 contaminant entities and respect the property's privacy-first, lightweight character throughout.

Governing constraint

No item in this plan introduces a third-party script, an analytics tag, a cookie, or an ad unit. Every content addition is statically rendered from the existing component library.

Week 1 — Days 1–7 · Stop the bleeding

Goal: nothing on the site misrepresents itself, and it can be contacted. No content work until this is done.

DAY	TASK	DETAIL & ACCEPTANCE
D1	Verify live origin	Issues 1, 13. Confirm canonical, robots.txt and header set against production. **Blocks everything else.**
D1	Brand decision	Issue 5. Choose Tap to Glass or waterqualitylens.com. Cannot be deferred — every later asset embeds it.
D2	Publish privacy, terms, contact	Issue 2. Copy is verbatim in §B; fill bracketed fields, set the date, wire into footer and sitemap.
D3	Execute the rename	Issue 5. Global replace, OG card, manifest, affiliate domain. Verify with `grep -ril`.
D3	In-context data disclosure	Issue 6. One line beneath the search field.
D4	Coverage honesty	Issue 4. The "no data ≠ clean water" empty state and hero expectation line. Highest trust-per-hour item in the plan.
D5	Fix manifest icons	Issue 7. Generate 192/512/maskable PNGs; remove the 404 entry.
D5	Lab-test CTA	Issue 11. Turns an existing honest argument into the first live revenue path.
D6	Rate limiting + CSP hash	Issues 8, 12. Both drop-in.
D7	Search Console & Bing	Register both, submit sitemap, capture the baseline indexed count against the expected 37 URLs.

Week 2 — Days 8–14 · Authority foundations

Goal: the site can demonstrate who stands behind it, and its best pages become eligible for rich results.

DAY	TASK	DETAIL & ACCEPTANCE
D8	E-E-A-T rewrite of /about	Issue 10. Named operator, background, funding model, corrections policy. Largest single ranking-signal gain available.
D9	Entity structured data	Issue 9. Dataset, DefinedTerm, ItemList. Validate every type in the Rich Results Test.
D10	Primary-source citations	Issue 23. Sources block on all 27 contaminant pages and /methodology.
D10	Freshness markers	Issue 24. Visible and machine-readable audit/sampling dates.
D11–12	**Content: "How to read your utility's water quality report"**	~1,800 words. The definitive CCR explainer — what utilities must publish, what the tables mean, what they legally may omit. Hub page; links to every contaminant entity. Targets a large, evergreen, genuinely underserved query set.
D13	**Content: "NSF 42 vs 53 vs 58 vs 401"**	~1,500 words. Extends the site's sharpest differentiator into a standalone asset. High commercial intent; links directly to /registry.
D14	A11y fixes	Issues 18, 19, 20. Focus trap, live region, radio semantics. Re-run an Axe pass.

Week 3 — Days 15–21 · Coverage and depth

Goal: begin closing the coverage gap and expand the indexable surface with genuinely useful pages.

DAY	TASK	DETAIL & ACCEPTANCE
D15–17	**Deploy the Worker & run ingestion**	Issue 3. Provision D1/KV/Queues, migrate, deploy, run the first SDWIS ingestion. The project's critical path.
D18	Expand PWS coverage	Target the 50 largest US systems by population. Converts /pws from a 3-row placeholder into a real index and gives the programmatic template fuel.
D19	**Content: "PFAS in US drinking water"**	~2,000 words. The highest-growth query cluster in the niche. Covers the 2024 federal limits, what they mean, and which certifications actually remove PFAS. Links to PFOA/PFOS/GenX/PFHxS/PFNA entities.
D20	**Content: "Does my home have a lead service line?"**	~1,400 words. High-intent, directly monetizable via lab tests, and pairs with the plumbing disclaimer the site already makes.
D21	Frontend test suite	Issue 17. Lock the set-cover safety invariants before coverage growth multiplies the blast radius.

Week 4 — Days 22–30 · Compounding assets

Goal: build the owned audience channel and the recurring-revenue mechanism; consolidate.

DAY	TASK	DETAIL & ACCEPTANCE
D22–23	Expose saved addresses & alerts	Surface the Worker's existing alert capability. Privacy-respecting opt-in, double opt-in email, one-click unsubscribe. The only owned-audience channel available.
D24	Cartridge reminder flow	Capture install date, schedule replacement email with affiliate link. Converts one-time purchases into recurring revenue.
D25	**Content: "Water filter pitcher comparison"**	~1,600 words. Commercial-intent, built directly from the certification registry so it cannot drift from the data. Highest conversion content on the site.
D26	**Content: "Nitrate in well and rural water"**	~1,300 words. Underserved, high-stakes (infant methemoglobinemia), and RO-exclusive — the nitrate warning already makes the technical argument.
D27	Internal linking pass	Every contaminant entity → its guide; every guide → the lookup; every PWS page → its detected contaminants. Wire the hub-and-spoke properly.
D28	Performance & medium issues	Issues 15, 25, 16. Suspense boundary on /results, CSS audit, begin retiring duplicated logic.
D29	Low-priority sweep	Issues 26, 27, 30, 31. All quick wins; batch in one commit.
D30	Measure & re-audit	Lighthouse on 5 representative routes; Search Console indexed count vs. Day-7 baseline; confirm all 6 critical issues closed.

Content plan summary

NEW ASSET	WORDS	PRIMARY INTENT
How to read your utility's water quality report	~1,800	Informational hub · lab test
NSF 42 vs 53 vs 58 vs 401	~1,500	Commercial investigation · hardware
PFAS in US drinking water	~2,000	Informational · highest growth
Does my home have a lead service line?	~1,400	High intent · lab test
Water filter pitcher comparison	~1,600	Transactional · hardware
Nitrate in well and rural water	~1,300	Informational · RO hardware
Total new editorial	**~9,600**	**Plus ~50 new PWS entity pages**

Why this sequence and not a faster one

It would be possible to front-load content and chase traffic in week 1. That would be a mistake here. Until the canonical origin is verified, published content may accrue to another domain. Until a contact page and privacy policy exist, YMYL content is competing with a structural E-E-A-T handicap. And until the coverage message is honest, every new visitor the content attracts meets the site's worst failure mode. Trust infrastructure first is slower for two weeks and faster for everything after.
